

Oumaima El Ghalbouni

Artificial Intelligence | Data Engineering | Data Science

Seeking a PFE Internship in Data Engineering / Artificial Intelligence – 2027

+212 616869644 | elghalbounioumama@gmail.com | AI Hoceima, Morocco

[linkedin.com/in/el-ghalbouni-oumama-a73a26331](https://www.linkedin.com/in/el-ghalbouni-oumama-a73a26331) | github.com/elghalbouni-oumama | Portfolio

Profile

Final-year Data Engineering engineering student (BAC+5) at ENSA AI Hoceima, with academic and professional experience in Artificial Intelligence, Data Science, Data Engineering, and application development. Hands-on experience in Machine Learning, NLP, RAG, and data processing. Seeking a PFE internship starting January 2027 in Artificial Intelligence/Data Engineering.

Professional Experience

AI & Data Science Intern, **Kenitra, Morocco**
Moroccan Center for Polytechnic Research and Innovation **07/2026 – 08/2026**

- Developed a phishing URL detection pipeline combining **Machine Learning, Elasticsearch, and Streamlit**; evaluated 4 models using 5-fold cross-validation, achieving **90%+ precision and recall**.
- Developed a 5-page Streamlit dashboard with **automated email alerts** and integrated threat intelligence feeds into Elasticsearch.

Data Analyst Intern, **AI Hoceima, Morocco**
Terra Nova EBS (France) **07/2025 – 08/2025**

- Designed a **4-page Power BI dashboard**, including DAX measures and calculated columns to monitor KPIs for the Procurement Department.
- Cleaned, transformed, and structured datasets using **Excel and Power BI**.

Web Development Intern, **AI Hoceima, Morocco**
CH Provincial Med VI **06/2025 – 09/2025**

- Designed and developed Administrator and Employee modules for a web application aimed at digitalizing the radiology department's operations.
- Contributed to database design, interface integration, and the **local deployment** of the application.

Projects

AI Meeting Assistant – NLP, RAG & Intelligent Assistant **06/2026**
[GitHub Project](#)

- Developed an intelligent application to analyze meeting transcripts and generate summaries and action plans.
- Implemented sentiment analysis, action item extraction, and a **FAISS + RAG** system for document-based question answering.

Technologies: Python, HuggingFace Transformers, FAISS, RAG, NLP, Streamlit

End-to-End SQL Data Warehouse & Churn Prediction **08/2026**
[GitHub Project](#)

- Designed a **Bronze/Silver/Gold** data warehouse in SQL Server with 19 automated data quality checks; developed a churn model achieving **97% recall and 0.98 ROC-AUC** on 18,482 customers.
- Deployed inference through **FastAPI** and Streamlit with a 5-page Power BI dashboard.

Technologies: Python, SQL Server, T-SQL, FastAPI, Power BI

TaaSim Casablanca – Real-Time Urban Mobility Platform **03/2026**
[GitHub Project](#)

- Designed a Kappa architecture using **Apache Flink and Spark** for real-time mobility data processing and integrated an ML model for demand forecasting.

Technologies: Kafka, PyFlink, Spark, Cassandra, MinIO, FastAPI, Docker

FHV Trip Data Optimization – MongoDB **03/2026**
[GitHub Project](#)

- Optimized MongoDB indexing and sharding on **16 million records**, reducing query execution time from **61.8s to 7.8s**.

Technologies: MongoDB, Python, Dash

Education

Engineering Degree – Data Engineering, <i>National School of Applied Sciences (ENSA) of AI Hoceima</i>	AI Hoceima, Morocco <i>2024 – Present</i>
Training in Data Engineering, Big Data, Artificial Intelligence, Machine Learning, and Business Intelligence.	
Preparatory Cycle, <i>National School of Applied Sciences (ENSA) of AI Hoceima</i>	AI Hoceima, Morocco <i>2022 – 2024</i>
Baccalaureate – Physics & Chemistry, French Option, <i>Rif High School</i>	Ajdir, Morocco <i>2022</i>

Skills

- **AI & Data Science:** Python, Scikit-learn, Pandas, NumPy, NLP, RAG
- **Data Engineering & Big Data:** ETL/ELT, Data Warehouse, Data Lake, HDFS, Avro, Parquet
- **Databases:** SQL, SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, Cassandra
- **Development & APIs:** Java, PHP, HTML, CSS, FastAPI, Spring Boot, REST APIs
- **BI & Tools:** Power BI, Data Modeling, Docker, Git, GitHub

Online Courses & Certifications

- **Advanced SQL for Data Engineering** – 365 Data Science
- **Oracle Cloud Infrastructure 2025** – Oracle
- **Supervised Learning with Scikit-learn** – DataCamp

Languages

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| ▪ Tarifit: Native | ▪ French: Upper-Intermediate |
| ▪ Arabic: Fluent | ▪ English: Intermediate |